Kuan-Yu Wey 衛冠瑜

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Education

University of California, Los Angeles Sep. 2021 – Now

M.S. in Physics (on the Ph.D. program track, GPA: 4.00/4.00) Sep. 2021 – Dec. 2022

Ph.D. Candidate in Physics and Astronomy (Condensed Matter Experiment, GPA: 4.00/4.00) Advance to Candidacy in June 2024

- Advisor: Dr. Christopher Gutiérrez Expected Graduation: June 2027
- Research focus: Imaging and controlling microscopic charge density wave (CDW) in low-dimensional materials

National Taiwan University, NTU Sep. 2017 – Jan. 2021

B.S. in Physics (GPA: 4.25/4.30; Ranking: 3/60)

- Advisor: Dr. Ya-Ping Chiu (邱雅萍博士), Dr. Mei-Yin Chou (周美吟博士)
- Research focus (1): Superconducting Proximity Effect at Cuprate-Insulator Interface, advised by Dr. Ya-Ping Chiu
- Research focus (2): First-Principles Study on Stacking of Twisted-Bilayer TMD Materials, advised by Dr. Mei-Yin Chou

Skills

- **Material Surface Characterization:** specialize in scanning tunneling microscope (STM) and atomic force microscope (AFM)
- **Material Bulk Characterization**: X-ray diffraction, angle-resolved photoemission spectroscopy (ARPES), transport measurement
- **Vacuum & Cryogenics**: ultra-high vacuum (UHV) systems with liquid nitrogen & helium transfer and recovery
- **Sample Preparation:** UHV in situ annealing and sputtering, chemical/physical vapor deposition (CVD/PVD)
- **Software**: Python (instrument control, advanced image analysis), CAD (design customized sample plate with heating/electrical contacts/strain accessibility), MATLAB, Gwyddion, ImageJ, Illustrator, NANONIS (Scanning Probe Controller/Software)

Research Experience

- **Dynamics of Charge-Density Wave Using Scanning Tunneling Microscope @ UCLA** Nov.2022 – Present
  - Probing the atomic-scale dynamics of a quasi-1D CDW system by STM/STS with in situ electrical contacts
  - Design experiments across multi-instrument control and optimize scanning conditions of particular samples
- **Band Structures Characterization by ARPES @ Lawrence Berkeley National Laboratory** May 2021 – Present
  Using ARPES to measure single crystals/2D devices under large strain or external electric fields, cooperating with beam scientists
- **Internship for First-Principles Study @ Academia Sinica in Taiwan** July 2020 – Jan. 2021
  First-Principles Study (Density functional theory) on different stackings of twisted-bilayer transitional metal dichalcogenides (TMD) – awarded Chau-Ting Chang Summer Research Scholarship (outstanding performance in summer research)
- **Internship for STM Field Emission Resonance (FER) Study @ Academia Sinica in Taiwan** July 2019 – Aug. 2019
  Analyze FER on Ag(100) and Ag(111) to reveal the surface dipole effects, advised by Dr. Wei-Bin Su and Dr. Chia-Sheng Chang

Publication and Presentations

[1] K.-Y. Wey, M. Mandigo-Stoba, S. Brown, C. Gutiérrez. Visualizing and controlling the atomic-scale depinning of a charge density wave. (In preparation, drafting for *Science* Journal)

[2] A. G. Prado, M. Mandigo-Stoba, K.-Y. Wey, S. Nekarae, A. Enriquez-Ibarra, S. Bañuelos, A. Nguyen, and C. Gutierrez. PyAtoms: An interactive tool for simulating atomic scanning tunneling microscopy images of 2D materials, moiré systems and superlattices. (Submitted to *Review of Scientific Instruments*)

[3] J. Green, H. W. T. Morgan, M. Mandigo-Stoba, W. T. Laderer, K.-Y. Wey, A. G. Prado, C. Jozwiak, A. Bostwick, E. Rotenberg, C. Gutiérrez, A. N. Alexandrova, and N. Ni. "Mapping the three-dimensional fermiology of the triangular lattice magnet EuAg4Sb". (*Physical Review B* **111**, 085139, 2025)

Three Conference Orals: (1) *IBS Conference on STM'25* @ Seoul, Korea. (2) 2025 *APS March Meeting* @ Anaheim, CA, USA. (3) 2024 *APS March Meeting* @ Minneapolis, MN, USA.

Two Conference Posters: (1) 2023 *CA-USG Workshop on 2D Materials* **(Best Poster Award)** @ Irvine, CA, USA. (2) 2020 *Physical Society of Taiwan* **(Excellent Poster Award, Top 1 of 26)** @ Pingtung, Taiwan

Awards

- Government Scholarship to Study Abroad (GSSA) from Taiwan June 2025 to May 2027
- Julian Schwinger Fellowship - granted to outstanding UCLA Physics students (<1 per year) Oct. 2021 to Sep. 2025
- Chau-Ting Chang Summer Research Scholarship, Academia Sinica Aug. 2020
- Presidential Award (Five Times), Department of Physics, NTU 2018 Spring & Fall, 2019 Spring & Fall, 2020 Fall